

# ZONGYAO YANG

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## SUMMARY

- **Full-stack Software Engineer** experienced in building and operating production web applications and data-driven systems using JavaScript, PHP, Python, and SQL.
- Delivered end-to-end features across frontend, backend, authentication, APIs, and automated data pipelines in distributed and research computing environments.
- Strong background in scalable web platforms, multi-site systems, and data visualization workflows.
- Authorized to work in the U.S. **without** visa sponsorship.

## EXPERIENCE

### Software Engineer

#### National Center for Atmospheric Research (NCAR)

Aug 2024 - Feb 2026    Boulder, CO

##### Full-Stack Web Platform

- Designed and built a scalable full-stack web platform using **JavaScript (React)**, **PHP (Symfony)**, **Python (Django)**, and **SQL** to manage visiting researchers and scientific collaborators from U.S. and international institutions.
- Modeled and implemented relational database schemas spanning multiple tables for visitor profiles, affiliations, and scheduling, ensuring data integrity and efficient querying.
- Developed dynamic administrative interfaces and backend services with **Node.js** to support onboarding, search, and coordination workflows used by cross-lab program staff.
- Integrated **Single Sign-On (SSO)** via **SimpleSAMLphp** and implemented role-based authorization, including super-admin controls, to secure access across institutional users.
- Optimized queries and application performance to support concurrent users and reliable access to collaboration data across research groups, maintaining a production system used for international visitor coordination.
- Containerized applications with **Docker** and deployed them to **AWS cloud infrastructure**, improving portability, scalability, and deployment efficiency.

##### Multi-Site Publications Sync System (Drupal)

- Developed a **Drupal module** for multi-site data synchronization using scheduled cron jobs to ingest publications from external APIs across downstream lab sites.
- Designed service-layer architecture in **PHP** to fetch, validate, and persist paginated API data into normalized SQL tables for cross-site consumption.
- Implemented fault-tolerant ingestion workflows with logging, retries, and validation to handle API failures and data inconsistencies.
- Delivered reusable view blocks and caching strategies to render synchronized publication data efficiently across distributed web properties.

##### WRF-Chem Forecast Visualization Pipeline (HPC)

- Engineered an **HPC-based processing pipeline** on NCAR Derecho to transform daily WRF-Chem NetCDF forecast outputs into web-ready visualization products.
- Processed large multi-dimensional scientific datasets with **Python Libraries (Xarray, NetCDF4, Numpy)** across chemical species, model domains, and elevation levels to support operational forecast visualization.
- Developed interactive **JavaScript interfaces** enabling dynamic selection of forecast variables and resolutions for research users.

### Software Engineer

#### Charter Communications

Jan 2022 - Jan 2024    Denver, CO

##### Application Development (Full Stack / Front End)

## PROGRAMMING LANGUAGES

JavaScript / TypeScript  
PHP    Python    SQL  
Fortran / MATLAB

## WEB & APPLICATION DEVELOPMENT

Angular    React    Redux  
NgRx    Node.js/Express  
Symfony    RxJS  
REST APIs    GraphQL  
Twig    HTML5  
CSS3 / SCSS    Drupal CMS

## BACKEND, DATA & SYSTEMS

Relational Databases  
NoSQL    Data Pipelines  
Large-Scale Datasets  
Authentication    SSO  
SAML    Access Control

## DEVOPS & PLATFORMS

Docker    Podman    Linux

## ENGINEERING PRACTICES

HPC Data Pipeline  
Git Version Control  
Production Debugging  
Log Analysis  
CI/CD Concepts    Jasmine  
Jest    Automated Testing

## AI DEVELOPER TOOLS

Claude    Codex    Cursor

- **Developed and maintained customer-facing web applications** on **spectrum.net**, supporting **high-traffic, authenticated user workflows** for eligibility checks, personalized product offers, and online purchases.
- **Built and enhanced modular, reusable UI components** using **Angular, SCSS** and **UI Libraries**, enabling consistent feature development across multiple product areas.
- **Integrated frontend applications with backend services** using **GraphQL**, improving data-fetch efficiency, API contract clarity, and long-term maintainability.
- **Worked with distributed backend systems** to consume and render dynamic data while handling partial failures, latency, and fallback states in the UI.
- **Implemented feature flags and configuration-driven behavior** to safely test, validate, and incrementally roll out new functionality while preserving **production stability**.
- **Applied accessibility best practices** by implementing **WCAG-compliant interfaces**, including semantic HTML, ARIA roles, keyboard navigation, and screen-reader support.
- **Collaborated with product and design teams** to translate business requirements into scalable, maintainable front-end implementations.
- **Participated in release and deployment workflows**, including branch creation, version tagging, and coordinated releases across **UAT, staging, and production environments**.

*Production Reliability, Validation, and Engineering Support*

- **Validated application behavior during development** through data verification, environment checks, and log analysis to ensure correctness prior to production deployment.
- **Applied shift-left engineering practices** by running **BDD-style workflows using Cucumber** to validate critical user journeys before formal QA handoff.
- **Partnered closely with QA engineers** to improve test coverage, reduce regressions, and accelerate release cycles.
- **Diagnosed and resolved UI and data-integration issues** across multiple environments by tracing requests through frontend, API, and backend layers.
- **Assisted with analytics instrumentation and verification**, ensuring accurate event tracking and performance behavior in production.
- **Supported post-release monitoring and issue resolution**, contributing to improved system reliability and reduced production defects over time.

Research Assistant (Scientific Software Developer)

Florida International University

 Aug 2014 - Dec 2020  Miami, FL

- **Developed and maintained scientific software systems** supporting hurricane forecasting and atmospheric modeling applications.
- Implemented and optimized numerical modeling workflows using **Fortran, MATLAB, and Python**, improving computational efficiency and large-scale simulation performance for hurricane intensification studies.
- **Built Python-based data processing and visualization pipelines** to analyze and present **large, multi-dimensional datasets** (NetCDF, GRIB).
- Built Python-based scientific data processing and visualization pipelines leveraging **xarray, NumPy, SciPy, Pandas, Matplotlib**, and **Cartopy** to analyze large multi-dimensional atmospheric datasets in **NetCDF** and **GRIB** formats.
- **Integrated modeling outputs into web-based applications** and relational databases, enabling accessible delivery of computational results.
- **Authored SQL queries and database schemas** to support efficient data storage, retrieval, and reporting.

EDUCATION

Ph.D in Earth Systems Science (computational focus)

Florida International University

 Aug 2014 - Dec 2020  Miami, FL

M.S. in Environmental Studies (Atmospheric Sciences)

University of Massachusetts Lowell (UMass Lowell)

 Aug 2012 - May 2014  Lowell, MA

B.S. in Computer Sciences

Nanjing University of Information Sciences & Technology

 Sep 2008 - July 2012  Nanjing, China

HOBBIES

 Basketball  SnowBoarding  Lure Fishing  Tennis  Swimming